

Electrical Design Description

Detailed electrical design of Boaty Mk1: power switching, schematics, custom boards, harness and calculations, for the Critical Design Review

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Document control

Issue	Date	Change	By
A	30 September 2026	First issue, draft for the CDR.	Claude (drafted)

Role	Name	Signature / date
Author	Claude, for Project Boaty	
Reviewer / approver	Project owner	

Review with the electrical zip open: the schematics PDF, the wire list and the board layouts. Section 2 has the decisions; section 8 every check; section 9 the change requests.

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1. Introduction

1.1 Purpose and status

This document describes the detailed electrical design of Boaty Mk1: power distribution and switching, the flight controller's wiring, the mission computer and its sensors, the mast and pod connections, and the Mission Control panel. It gives the schematics, the three custom boards, the harness and the calculations behind them, and checks the design against the requirements allocated to it. With the Mechanical Design Description it is an input to the Critical Design Review.

It is a **draft for the CDR**. The system documents stay as baselined at the SDR (tag sdr-baseline-1); where this design cannot meet one of them, section 9 proposes a numbered change request.

1.2 How the design is held and checked

The design is one Python file, `electrical/boaty_elec/design.py`: every part with its pins, every net, which connector halves mate, and every harness wire with its gauge, length and route. Everything else is generated from it and checked against it:

- an **electrical rule check**: every pin on exactly one net; every net one connected piece once boards, mated connectors and harness wires are counted, and nothing else joined; no input without a source; no pin above its maximum voltage; connector keying (fault-injected: it finds a missing wire, a pin on two nets, a wire on the wrong net, an over-voltage);
- the **schematics**: each drawn pin takes its net name from the netlist, and a wire can only be drawn between pins of one net, so a sheet cannot disagree with the netlist; a coverage check proves every pin is on a sheet;
- the **stripboard layouts**: rebuilt from copper strips, cuts and links alone and compared with the netlist (a missing link shows as an open, a missing cut as a short);
- the **calculations**, including a time-stepped simulation of the key switch and its bleeder, and the compliance matrix in section 8.

This issue was built from commit 82541bc. `python3 electrical/build_elec.py` rebuilds the schematics, netlist, wire list, BOM, layouts, figures and the results file this document reads.

1.3 References

Ref	Document
[1]	BOATY-SRS-001 Issue H System Requirements Specification
[2]	BOATY-ADD-001 Issue H Architecture Design Document
[3]	BOATY-ICD-001 Issue H Interface Control Document (IF-04 to IF-09, IF-12, IF-22)
[4]	Subsystem specifications: HUL B, PRP C, PWR D, HLM F, MCP F, MCN F, REC B, SIM G
[5]	BOATY-KCL-001 Issue D Key Component List
[6]	BOATY-MDD-001 Issue A Mechanical Design Description (draft)
[7]	ArduPilot SpeedyBeeF405WING board definition and README (docs/kcl/sources)

2. Summary

Quantity	Design	Requirement
Checks: pass / by test / not met	35 / 4 / 2	41 checks
ERC; board layouts; schematic coverage	0 errors; 0 errors; complete	0
Key-in inrush; key-out to < 0.5 V	5.6 A; 58 ms	soft; ≤ 100 ms (IF-06)
ESC input at the power limit (critical battery)	7.17 V at 70 W; 7.63 V at 60 W	≥ 7.5 V (PWR-D22)
Cruise load; endurance	11.34 W; 114 min	≥ 40 min
Box air, 30 °C in sun	44 °C	≤ 50 °C

Quantity	Design	Requirement
Bill of materials	£229 (£246 buying everything)	≤ £185

Decisions for the owner at the CDR.

- P-06 PWR-D22, HLM-D43, KF-02: Every supply works at the rail voltage the power limit allows (critical battery, acceptance-limit cells): ESC input 7.17 V at 70 W (harness 34 mΩ); 7.5 V needs ≤ 63 W → **CR-12**
- X-01 CON-001, CR-08: Bill of materials within the £185 cap (boat and shore kit): £246 buying everything; £229 with workshop stock (wire, resistors, stripboard) → **CR-14**
- CR-13 (no failing check): the ICD's key-switch circuit (gate RC soft start, no discharge) does not meet IF-06's own test; this design changes the circuit.

What this design adds to the system, in one line each:

- a **power and interconnect board** (PIB): a solid-state main switch driven by a small IP67 toggle, the key switch with a gate-drain soft start and an active rail bleeder, the rail sense divider, surge suppressors, a PTC on the buck feed, and the mast-cable junction;
- a **mission-computer interface board** (MIB) on the Pi Zero's header: protected 5 V entry, UART, DS18B20 and moisture inputs;
- a **panel board** (PNL) for Mission Control: four button inputs and four LED drivers on the IF-12 GPIOs;
- an **M12 8-pole mast connector** at the handle so the mast comes off without opening the box, and **IP68 pod connectors** on the stern decks (TBC-12 still open);
- the **harness**: 16 AWG main path, 18 AWG ESC leads, four cables through the four PG7 glands.

3. Architecture

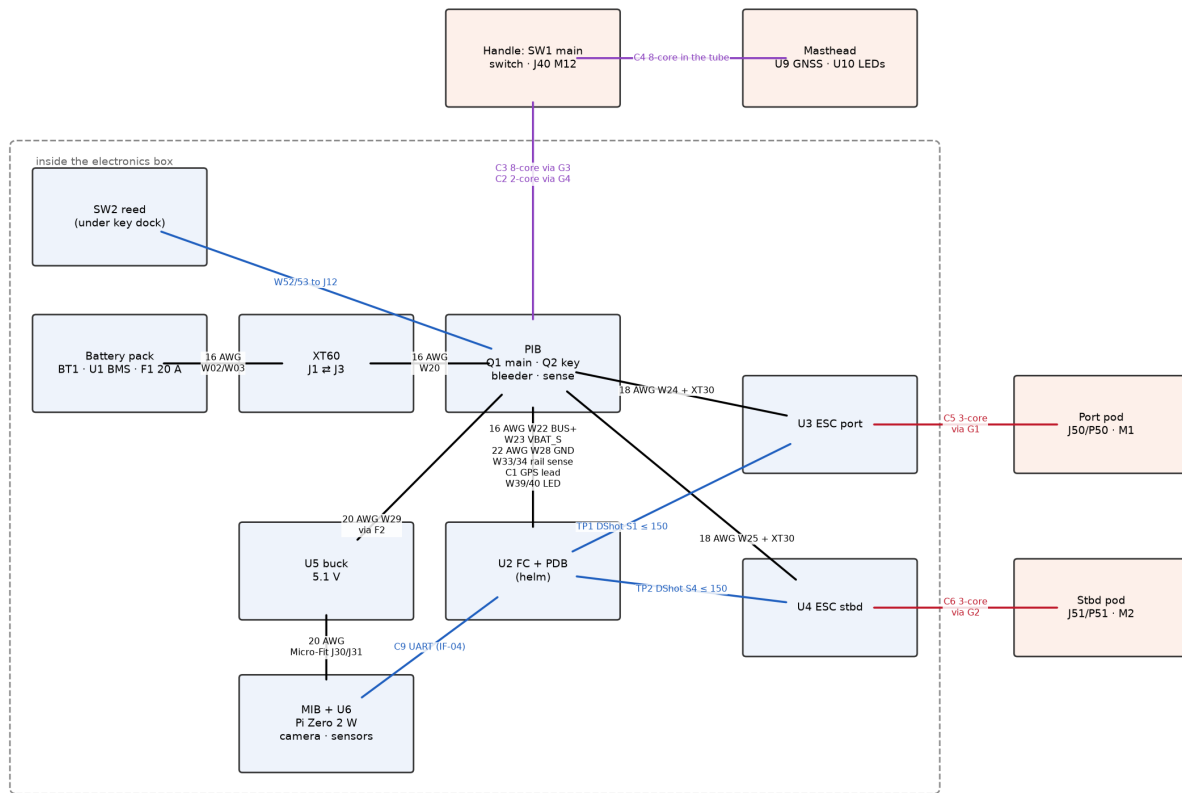


Figure 1. Boards, modules and cables. Blue: signals; black: power; purple: mast and switch cables; red: motor cables through the glands.

3.1 Power path

Battery (3 × 18650 and BMS, PWR-D01/D02) → 20 A fuse within 40 mm of the BMS output → **Q1**, the main switch on the PIB → the PDB's battery pads and current sensor → the PDB's battery output (VBAT_S). From VBAT_S: the flight controller's own regulator and the 5.1 V buck (both before the key, PWR-D10/D11), and **Q2**, the key switch → the motor rail → two XT30 leads to the ESCs. Every load sits behind the PDB sensor, so the helm's mAh count includes the mission computer (IF-07).

Q1 and Q2 are high-side P-MOSFETs (PWR-D07 forbids low-side switching). The main switch is the solid-state option of PWR-D06: a small sealed IP67 toggle with a flip cover on the handle pulls Q1's gate down; the toggle carries a milliamp, so it needs no current rating and its lead goes through the fourth gland.

3.2 Grounds

One ground, star-connected at the PDB. Motor current returns from each ESC to the PDB on its own 18 AWG lead; the PIB's ground is a 22 AWG signal return that carries no motor current; the ESC signal grounds are twisted with their signals (IF-05). Mission Control is a separate system with its own ground (PI5_GND) and no wire to the boat.

3.3 Connectors and keying (PWR-D14)

Function	Connector	Keying
Battery	XT60 (sockets on the pack)	only XT60
ESC supply ×2	XT30	only XT30; same job twice
Balance (charger only)	JST-XH 4	only JST-XH
Mission computer 5 V	Molex Micro-Fit 2	only Micro-Fit

Function	Connector	Keying
Board signals	Molex KK 254 (2.54 mm, polarised)	each board uses each pin count once (PIB 2, 3, 6, 8; MIB 2, 3, 4); pin 2 of J12 and pin 4 of J34 are left empty
Mast (GNSS, compass, LEDs)	M12 A-coded 8-pole, IP67	coded; one per boat
Pods ×2	3-pole IP68 (TBC-12)	keyed; same job twice

JST-PH (2.0 mm) was the first choice for board signals; it does not sit on 0.1-inch stripboard, and JST-XH is the balance lead's connector, so both were ruled out.

3.4 Cable entries (MEC-012)

Gland	Size	Cable
G1	PG7	Port motor cable C5 (Ø6)
G2	PG7	Starboard motor cable C6 (Ø6)
G3	PG7	Mast cable C3 (8-core Ø5.5)
G4	PG7	Main switch cable C2 (2-core Ø4)

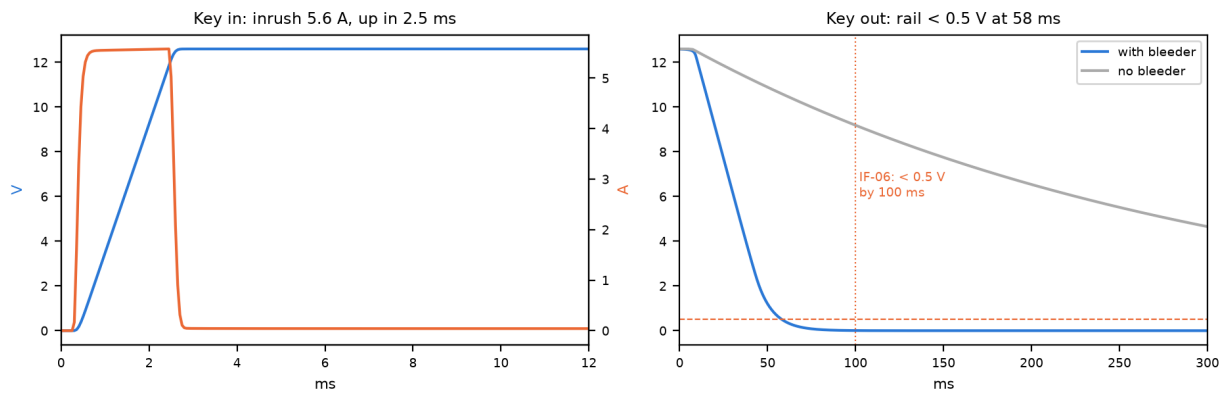


Figure 2. Key switch, simulated: key in (left) and key out with and without the bleeder (right).

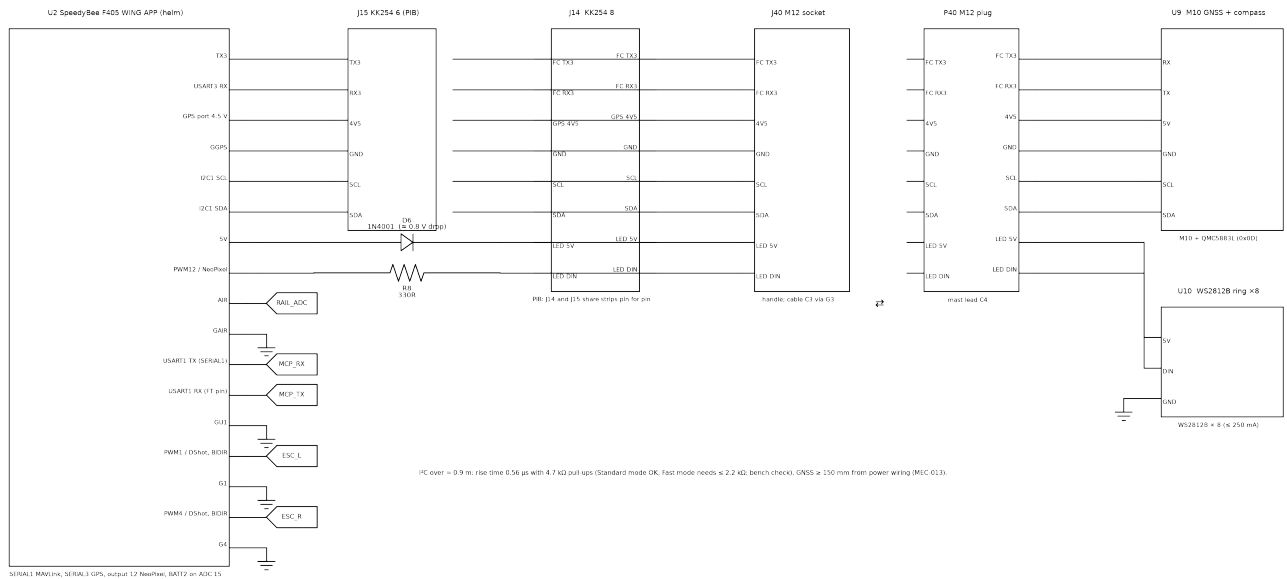
A gate-to-source capacitor (the ICD's 'gate RC ≈ 10 ms') does not soft-start a low- $R_{DS(on)}$ MOSFET: the simulation gives a 24 A inrush in 2.6 ms, because the FET turns fully on within a volt of threshold. The gate-to-drain capacitor holds the gate on its Miller plateau while the rail rises, which limits the inrush to 5.6 A. Without the bleeder the ESCs' capacitors would hold the rail up for 956 ms against IF-06's 100 ms; with it, 58 ms. The bleeder conducts for ≈ 237 ms and then stops, so a Q2 that has failed short (FM-22) cannot burn it out; the checklist's rail test (MCN-D54) finds that failure. The main switch Q1 soft-starts the bus the same way (6.1 A).

Ref	Value	Description / part	Source
Q1	P-MOSFET	Main switch, high side (PWR-D06 solid-state option) candidate Infineon IPP80P03P4L-04 (-30 V, $R_{DS(on)} \leq 6$ m Ω at -10 V assumed); fallback IRF4905 (20 m Ω)	EST
Q2	P-MOSFET	Motor-rail key switch, high side (PWR-D07, IF-06) candidate Infineon IPP80P03P4L-04 (-30 V, $R_{DS(on)} \leq 6$ m Ω at -10 V assumed); fallback IRF4905 (20 m Ω)	EST
Q3	IRLZ44N	Rail bleeder, logic-level N-MOSFET IRLZ44NPbF	DS
R1	10k	Resistor 10k, 0.25 W, 1%	CALC
R2	3k3	Resistor 3k3, 0.25 W, 1%	CALC
C1	470nF	Capacitor 470nF, 25 V, ceramic/film	CALC
D1	BZX85C15	Zener 15 V 1.3 W, Q1 gate clamp	DS
D2	P6KE18A	TVS 600 W, 15.3 V stand-off, main bus	DS
R3	10k	Resistor 10k, 0.25 W, 1%	CALC
R4	3k3	Resistor 3k3, 0.25 W, 1%	CALC
C2	470nF	Capacitor 470nF, 25 V, ceramic/film	CALC
D3	BZX85C15	Zener 15 V, Q2 gate clamp	DS
C3	1 μ F	Capacitor 1 μ F, 25 V, ceramic/film	CALC
R5	100k	Resistor 100k, 0.25 W, 1%	CALC
D4	BZX85C10	Zener 10 V, Q3 gate clamp	DS
R9	10R	Resistor 10R, 2.0 W, 1%	CALC
D5	P6KE18A	TVS 600 W, motor rail (ESC regeneration)	DS
R6	10k	Resistor 10k, 0.25 W, 1%	CALC
R7	1k	Resistor 1k, 0.25 W, 1%	CALC
C4	100nF	Capacitor 100nF, 16 V, ceramic/film	CALC
F2	PTC 1.1 A	Resettable fuse, buck feed RXEF110	DS
D6	1N4001	LED ring supply drop: 5 V $\rightarrow \approx 4.3$ V so 3.3 V data is a valid high (closes TBC-08)	DS

Ref	Value	Description / part	Source
R8	330R	Resistor 330R, 0.25 W, 1%	CALC
J11	KK254 2	Main switch lead Molex 22-27-2021	DS
J12	KK254 3	Reed switch lead (pin 2 not fitted: keyed against J11) Molex 22-27-2031	DS
J14	KK254 8	Mast cable from the gland Molex 22-27-2081	DS
J15	KK254 6	GNSS lead to the FC GPS port Molex 22-27-2061	DS

S3 Helm, GNSS and beacon (IF-04, IF-05, IF-06, IF-09, IF-22)

S3 Helm, GNSS and beacon (IF-04, IF-05, IF-06, IF-09, IF-22)



Sheet S3.

The helm's signal connections. GNSS and compass go from the FC's GPS port through J15 and J14 on the PIB (the two headers share strips pin for pin), the M12 socket on the handle and the mast lead. The LED data passes R8 (330 Ω); the ring's supply passes D6, which lowers it to ≈ 4.3 V so the FC's 3.3 V data is a valid high (section 5.5).

S4 Propulsion: ESC signals and motor phases (IF-05, IF-17, PRP-3)

ESC signals from outputs 1 and 4 (BIDIR-capable, KCL KF-09) as twisted pairs; motor phases through the glands and the deck connectors.

S5 Mission computer and its interface board (MIB): IF-03, IF-04, IF-08, MCP-D03/D26

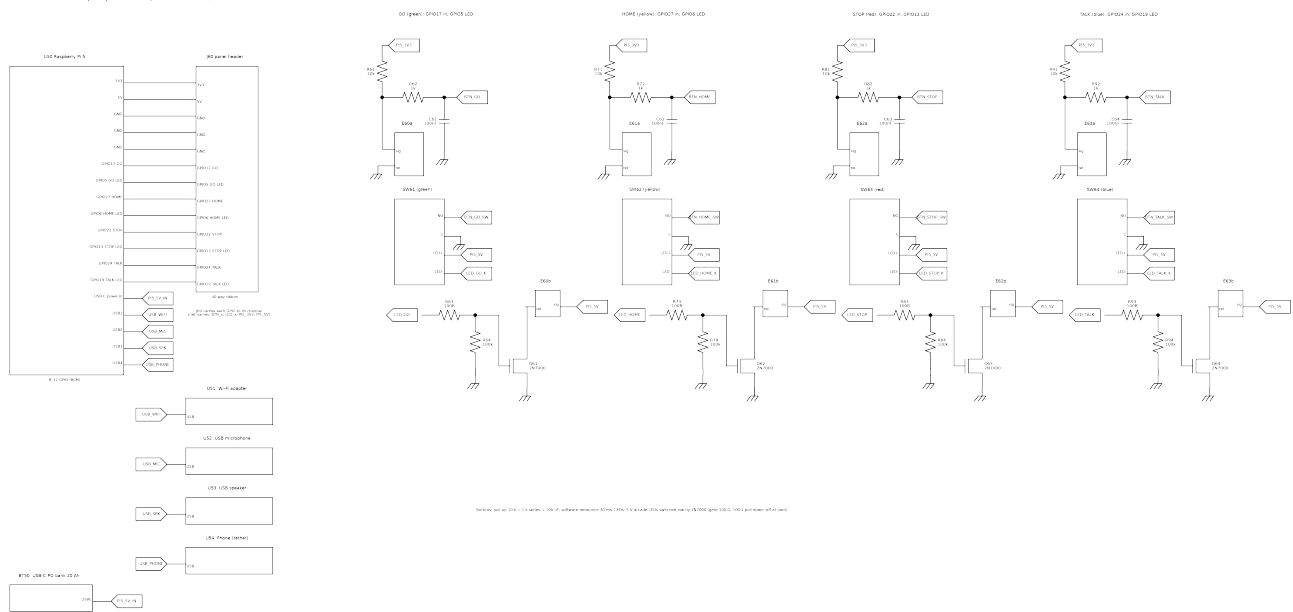


The MIB on the Pi Zero header: the 5 V feed enters on pins 2 and 4 behind a 5.8 V TVS and 470 μ F (back-powering bypasses the Pi's own protection); UART on pins 8/10; DS18B20 on GPIO4; the moisture comb on GPIO17 through 1 k with a 100 k pull-up and 100 nF filter. GPIO numbers match the software (sensors.py).

Ref	Value	Description / part	Source
J31	Micro-Fit 2 header	5 V in Molex 43650-0200	DS
D10	P6KE6.8A	TVS 5.8 V stand-off on the 5 V feed	DS
C10	470µF	Capacitor 470µF, 10 V, low-ESR electrolytic	CALC
C11	10µF	Capacitor 10µF, 16 V, ceramic/film	CALC
J32	2×20 socket	40-way socket onto the Pi Zero header (used pins only listed) 2×20 2.54 mm female, stacking	EST
J33	KK254 3	UART to the helm (IF-04) Molex 22-27-2031	DS
J34	KK254 4	DS18B20 lead (pin 4 not fitted: keyed against J33) Molex 22-27-2041	DS
J35	KK254 2	Moisture comb lead Molex 22-27-2021	DS
R10	4k7	Resistor 4k7, 0.25 W, 1%	CALC
R11	1k	Resistor 1k, 0.25 W, 1%	CALC
R12	100k	Resistor 100k, 0.25 W, 1%	CALC
C12	100nF	Capacitor 100nF, 16 V, ceramic/film	CALC

S6 Mission Control: Raspberry Pi 5 and the panel board (IF-12, MCN-1)

S6: Mission Control: Raspberry Pi 5 and the panel board (IF-12, MCN-1)



Sheet S6.

Mission Control. Each button input has a 10 k pull-up, 1 k series resistor and 100 nF; each 5 V button LED is switched low by a 2N7000 with a 100 k gate pull-down so the LEDs are off while the Pi boots. The GPIOs are those in IF-12 and in the software (panel.py); section 8 checks they agree.

Ref	Value	Description / part	Source
J60	2×20 IDC header	Panel board header, 150 mm ribbon to the Pi 5	EST
R61	10k	Resistor 10k, 0.25 W, 1%	CALC
R62	1k	Resistor 1k, 0.25 W, 1%	CALC
C61	100nF	Capacitor 100nF, 16 V, ceramic/film	CALC
Q61	2N7000	GO LED driver	DS

Ref	Value	Description / part	Source
R63	100R	Resistor 100R, 0.25 W, 1%	CALC
R64	100k	Resistor 100k, 0.25 W, 1%	CALC
R71	10k	Resistor 10k, 0.25 W, 1%	CALC
R72	1k	Resistor 1k, 0.25 W, 1%	CALC
C62	100nF	Capacitor 100nF, 16 V, ceramic/film	CALC
Q62	2N7000	HOME LED driver	DS
R73	100R	Resistor 100R, 0.25 W, 1%	CALC
R74	100k	Resistor 100k, 0.25 W, 1%	CALC
R81	10k	Resistor 10k, 0.25 W, 1%	CALC
R82	1k	Resistor 1k, 0.25 W, 1%	CALC
C63	100nF	Capacitor 100nF, 16 V, ceramic/film	CALC
Q63	2N7000	STOP LED driver	DS
R83	100R	Resistor 100R, 0.25 W, 1%	CALC
R84	100k	Resistor 100k, 0.25 W, 1%	CALC
R91	10k	Resistor 10k, 0.25 W, 1%	CALC
R92	1k	Resistor 1k, 0.25 W, 1%	CALC
C64	100nF	Capacitor 100nF, 16 V, ceramic/film	CALC
Q64	2N7000	TALK LED driver	DS
R93	100R	Resistor 100R, 0.25 W, 1%	CALC
R94	100k	Resistor 100k, 0.25 W, 1%	CALC

5. Calculations

5.1 Supply sag at the power limit (PWR-D22, KF-01/02)

The SDR sized the helm's power limit (BATT_WATT_MAX 70 W) so the rail stays at 7.5 V at the critical battery with acceptance-limit cells (0.22 Ω pack), with no allowance for the harness. The limit acts on the power the PDB measures, so the operating point solves $V_{PDB} \cdot I = P$ with $V_{PDB} = 9.6 \text{ V} - I \cdot (R_{\text{pack}} + R_{\text{up}})$; the ESCs see a further $I \cdot R_{\text{down}}$.

Series element	m Ω
fuse F1	3.5
BMS lead W01+W02	1.8
XT60	0.5
box lead W20	1.6
Q1 main switch	6.0
PIB to PDB W22	1.3
negative W03+W21	3.6
PDB shunt	0.3
battery to PDB	18.6
PDB to PIB W23	1.3
Q2 key switch	6.0
ESC lead W24 (+)	3.8
ESC lead W26 (–)	3.8
XT30	1.0
PDB to ESC	15.9

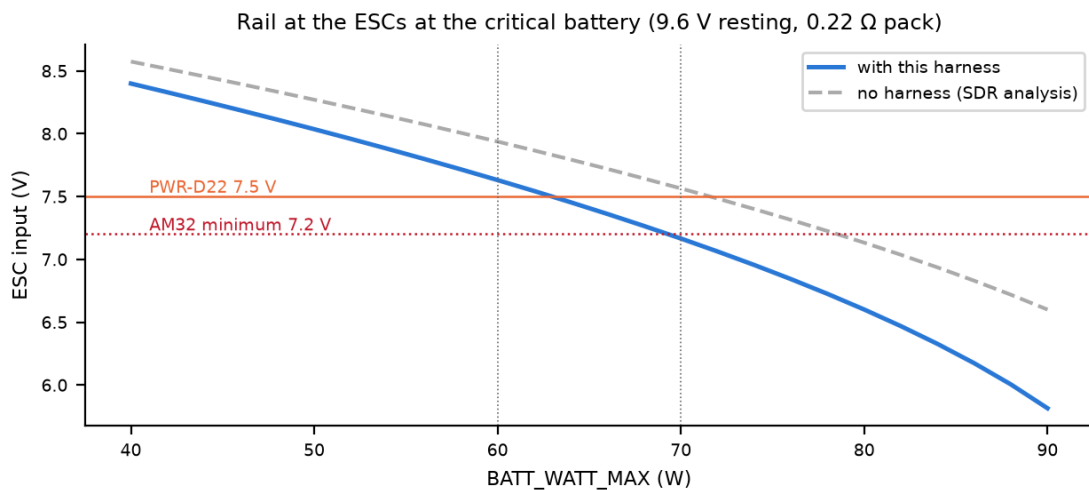


Figure 3. ESC input voltage against the power limit.

Case	Current	V at PDB	V at ESC
70 W, no harness (SDR)	9.3 A	7.56	7.56
70 W, this harness	9.6 A	7.32	7.17
70 W, fallback MOSFETs (20 m Ω)	9.8 A	7.11	6.82
60 W, this harness	7.7 A	7.75	7.63

With the harness the ESCs see 7.17 V at 70 W: below PWR-D22's 7.5 V and just below the AM32's 7.2 V. Holding 7.5 V needs ≤ 63 W; 60 W gives 7.63 V and still 1.85 N of thrust against the 1.1 N ENV-002 needs (70 W: 2.06 N). CR-12 proposes 60 W. The fallback MOSFET (IRF4905) is worse still, so the ≤ 6

mΩ part matters.

5.2 Switch MOSFETs

V_{GS} at 12.6 / 9.0 / 7.5 V: -9.5 / -6.8 / -5.6 V (−20 V maximum; 15 V zener). At the power-limited 9.6 A, with $R_{DS(on)}$ taken $2.25 \times$ its −10 V rating for low gate drive and a hot junction, Q1 dissipates 1.24 W: junction 127 °C in free air in a 50 °C box, 78 °C with a clip-on heatsink (fitted). At cruise (≈ 1 A) the loss is negligible.

5.3 Fuses and wires (PWR-003, PWR-D15)

F1 20 A is $2.1 \times$ the power-limited current and $1.2 \times$ the hardware maximum (2×8 A, PRP-D05, plus avionics), within PWR-003's $2 \times$. The BMS (≤ 25 A, fast) covers faults the fuse clears slowly, including the 18 AWG ESC leads (16 A). F2 (PTC, 1.1 A hold) protects the 20 AWG buck feed; the buck draws ≤ 0.46 A. Every wire carries its continuous current within its rating; the worst drop is 48 mV.

5.4 Rail sense (IF-06, PWR-D08)

10 k / 1 k: $\times 11.0$ (BATT2_VOLT_MULT 11.0). Full charge 1.15 V, arming threshold 9.0 V $\rightarrow$ 0.82 V at the ADC (3.3 V full scale); 8.9 mV per count at the rail; 1.1 mA; source impedance 909 Ω; filter 91 μs.

5.5 LED data level (IF-09, TBC-08)

A WS2812B needs $V_{IH} \geq 0.7 \cdot V_{DD}$. At 5.2 V that is 3.64 V, above the FC's 3.3 V output. D6 (1N4001) drops the ring's supply:

BEC, load	V_{DD} ring	V_{IH}	Margin	Margin without D6
5.0 V, 50 mA	4.30	3.01	0.29	-0.20
5.0 V, 250 mA	4.15	2.91	0.39	-0.20
5.2 V, 50 mA	4.50	3.15	0.15	-0.34
5.2 V, 250 mA	4.35	3.05	0.25	-0.34

This closes TBC-08 by design; the bench confirms it with the real ring and cable.

5.6 Moisture input (MCP-D03, FS-010)

The Pi's ≈ 50 k pull-up and R12 (100 k) give 33 k; GPIO17 reads low below 10.7 kΩ across the comb. Pond water (≈ 300 μS/cm) 1 mm deep across the comb measures ≈ 82 Ω, so 0.8% of the comb wet trips it; a condensation film does not. Traces are only energised through 33 k, so corrosion matters only once the box is already wet.

5.7 I²C over the mast (IF-22)

0.9 m of cable, ≈ 140 pF. Rise time 0.56 μs with 4.7 kΩ pull-ups, 0.26 μs with 2.2 kΩ: fine at 100 kHz (≤ 1 μs), marginal at 400 kHz (≤ 0.3 μs) unless the module's pull-ups are 2.2 kΩ. The board definition sets no bus speed; the bench confirms compass reads over the full cable (TEST).

5.8 Power budget and thermal (PWR-009/010, PWR-D17/D19)

Load	W
Flight controller + GNSS	0.80
Pi Zero 2 W + camera	1.80
LED beacon (average)	0.30
ESC idle (2)	0.20
Motors at cruise	8.00
Buck loss (90%)	0.20
PIB: switches, gate dividers, sense	0.04
Total at cruise	11.34

Endurance 114 min on 21.6 Wh (≥ 40). The budget file holds 11.1 W; this design adds the buck and PIB losses (CR-12 updates it).

Box in sun at 30 °C: 3.6 W inside plus 7.5 W of sun on the lid (the DUPLO plate shades 47%), lost from 0.109 m² at 7 W/m²K: 44 °C against PWR-D19's 50 °C and B6's 60 °C RTL. Verified by V-10.

5.9 Mission Control supply

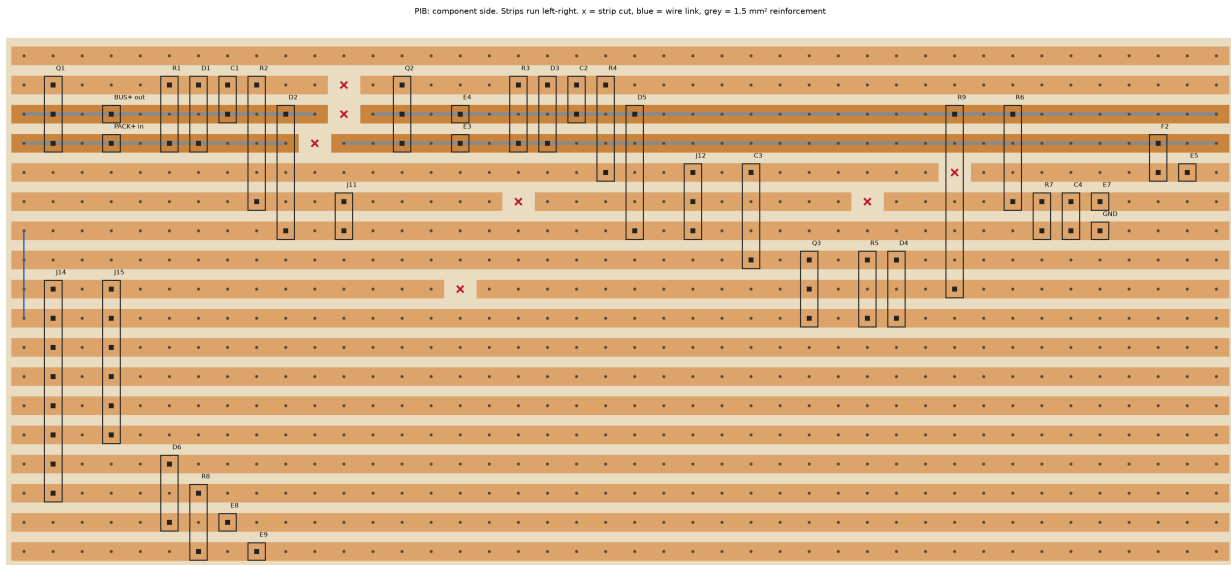
USB load	A
Wi-Fi adapter (TX)	0.45
USB speaker (peaks)	0.50
USB microphone	0.05
Phone on the tether (SDP)	0.50
Total (limit with usb_max_current_enable=1)	1.50 (≤ 1.6)

At 6.5 W the 20 Ah bank lasts 9.7 h (MC-001 ≥ 2.5 h).

6. Custom boards

All three are 0.1-inch stripboard with the strips running left to right in the drawings (component side). A red cross is a strip cut (a drilled hole, or a knife cut between adjacent holes); a blue line is an insulated wire link. The layout tool puts each part in its own column and plans the rows so most nets are a single strip, then cuts and links what is left, and the result is rebuilt from the copper and compared with the netlist.

6.1 PIB

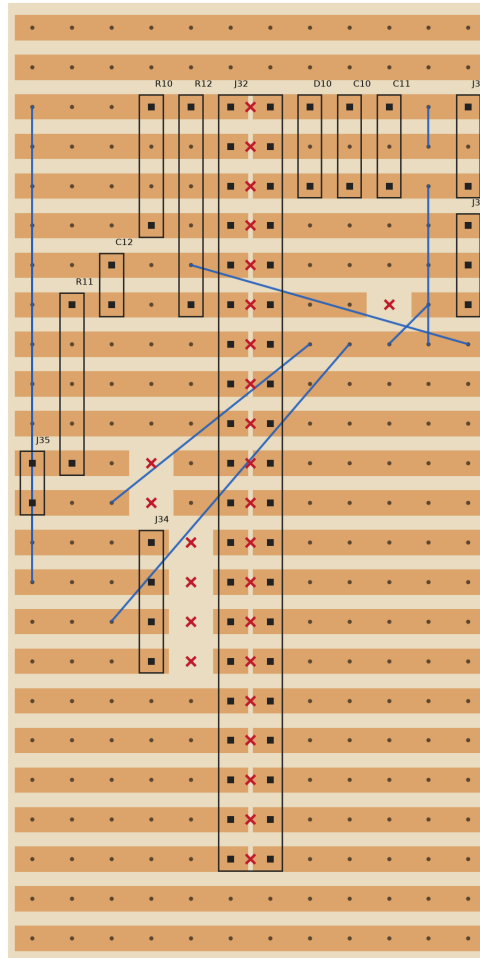


PIB: 42 × 18 holes (107 × 46 mm), 36 parts, 7 cuts (0 knife), 1 links; verified: 0 errors.

Power and interconnect board, 107 × 46 mm, on the mezzanine plate. The four high-current nets (PACK+, BUS+, VBAT_S, MOTOR+) are each one strip, reinforced with 1.5 mm² tinned copper soldered along it (grey), and the 16 AWG leads solder to pads E1-E4 on those strips. Q1 and Q2 lie flat with clip-on heatsinks.

6.2 MIB

MIB: component side. Strips run left-right. x = strip cut, blue = wire link, grey = 1.5 mm² reinforcement



MIB: 12 × 24 holes (30 × 61 mm), 12 parts, 27 cuts (20 knife), 8 links; verified: 0 errors.

Mission-computer interface board, 30 × 61 mm, the Pi Zero's footprint, on a stacking 2×20 socket. The header stands across the strips so odd pins feed left and even pins right (a knife cut between each pair).

6.3 PNL

Panel board, 112 × 61 mm, in the Mission Control case on a 40-way ribbon to the Pi 5; the buttons are wired to its pads.

7. Harness and packaging

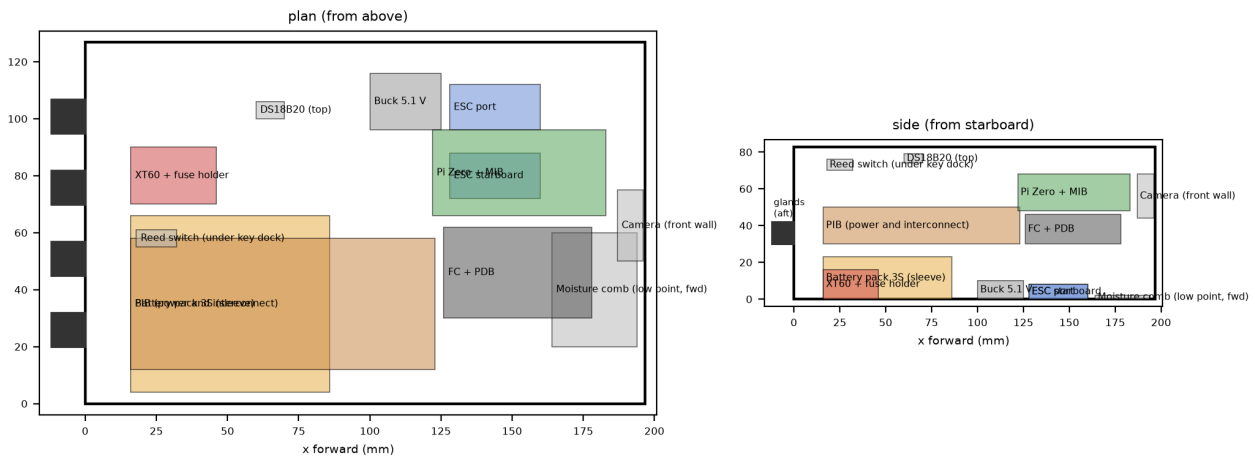


Figure 4. Inside the box. Floor: battery aft, ESCs and buck forward. Mezzanine plate at 30 mm: PIB aft, FC, Pi Zero and MIB forward at the camera window. The moisture comb sits at the front of the floor, the low point at design trim.

Battery to the nearest ESC: 42 mm ($PWR-D19 \geq 30$). No part overlaps another or the gland nuts. The reed switch sits on a bracket inside the aft wall, 15 mm under the lid's key dock, so the lid lifts off with nothing attached; the DS18B20 is at the top of the box (MCP-D26).

7.1 Wire list (summary)

Group	Gauge	Wires	Note
Pack leads	16 AWG	W01-W03	fuse ≤ 50 mm
Main path in the box	16 AWG	W20-W23	XT60 $\rightarrow$ PIB $\rightarrow$ PDB $\rightarrow$ PIB
ESC supplies	18 AWG	W24-W27	XT30 pairs
Buck feed and output	20 AWG	W29-W32	PTC; Micro-Fit
Signals in the box	22-26 AWG	W28, W33-W46, W52/53, W100-W107	twisted pairs for DShot and rail sense
Mast cable (box $\rightarrow$ handle)	$8 \times 0.25 \text{ mm}^2$	W60-W67	C3 via G3
Mast lead (handle $\rightarrow$ masthead)	$8 \times 0.25 \text{ mm}^2$	W70-W78	C4 in the tube
Motor cables	$3 \times 0.75 \text{ mm}^2$	W800-W812	C5/C6 via G1/G2 to the deck sockets
Pod leads	18 AWG	W900-W912	potted in the strut groove
Main switch lead	$2 \times 24 \text{ AWG}$	W50/W51	C2 via G4

The full list, with every wire's ends, colour and length, is wirelist.csv in the zip.

8. Compliance matrix

Computed by boaty_elec/checks.py at the commit on the cover. PASS: met. TEST: shown only on the bench or water; the test is named. FAIL: not met; the change request is named.

8.1 Design integrity

ID	Requirement	Check	Design value	Limit	Status
D-01	ERC	Netlist rule check (every pin placed, nets joined only by boards, mates and wires, drivers, voltages, keying)	0 errors; 120 parts, 70 nets, 114 wires	0 errors	PASS
D-02	Boards	Stripboard layouts rebuilt from copper, cuts and links, compared with the netlist	0 errors on 3 boards	0 errors	PASS
D-03	Schematics	Every part and pin on a sheet; every drawn wire joins pins of one net only if the sheets were built in this run	0 missing	0	PASS

8.2 Power

ID	Requirement	Check	Design value	Limit	Status
P-01	PWR-001, PWR-D01	One removable 3S Li-ion pack, 25-50 Wh	3S1P, 27 Wh nominal	25-50 Wh	PASS
P-02	PWR-002, PWR-D02	BMS: over-discharge, over-current, short circuit, balance confirm the bought board's trip points (bench)	OCP 25 A, ODV 2.7 V/cell	trip $\leq$ 25 A, 2.5-2.8 V	PASS
P-03	PWR-003, PWR-D05	Main fuse: position and rating the BMS (25 A, fast) protects the 18 AWG ESC leads above 16 A	20 A ATO, 40 mm lead (W01); 2.1 $\times$ the power-limited current; 1.2 $\times$ the hardware maximum (2 $\times$ 8 A)	$\leq$ 50 mm; $\leq$ 2 $\times$ max continuous	PASS
P-04	PWR-004, PWR-D06	Main switch waterproof, outside the box, without opening it	IP67 toggle with cover on the handle; drives Q1 (P-FET, 80 A class, solid-state option)	IP67; $\geq$ 20 A or a $\geq$ 30 A solid-state switch	PASS
P-05	PWR-005, PWR-D12	Separate 5 V for the avionics; no brown-out on a full thrust step at 9.6 V (resting) the FC and buck take their supply before the key switch (PWR-D10)	FC fed at the PDB: 7.32 V at 70 W	FC $\geq$ 7.0 V, buck $\geq$ 6.0 V	PASS
P-06	PWR-D22, HLM-D43, KF-02	Every supply works at the rail voltage the power limit allows (critical battery, acceptance-limit cells) at 60 W: 7.63 V and 1.85 N thrust ($\geq$ 1.1 N for ENV-002)	ESC input 7.17 V at 70 W (harness 34 m Ω); 7.5 V needs $\leq$ 63 W	$\geq$ 7.5 V (ESC rated 7.2 V)	FAIL CR-12
P-07	PWR-006, PWR-D13, IF-07	Battery V and I measured and logged bench calibration (KF-10)	PDB sensor $\rightarrow$ BATT_VOLT_PIN 10 / BATT_CURR_PIN 11; every load behind it	$\pm$ 1% V, $\pm$ 5% I after calibration	TEST
P-08	PWR-007, PWR-D04	No charge port; charged off the boat	balance lead J2 and XT60 only	no port	PASS
P-09	PWR-008, PWR-D14	Polarised connectors, one job per type	XT60 battery $\cdot$ XT30 ESCs $\cdot$ JST-XH balance $\cdot$ Micro-Fit Pi 5 V; signal KK254 with one pin count per board	no shared type	PASS

ID	Requirement	Check	Design value	Limit	Status
P-10	PWR-D15	Wire gauges and joints	16 AWG battery → bus, 18 AWG to the ESCs, silicone; soldered, heat-shrunk; glands on every entry	16 / 18 AWG	PASS
P-11	PWR-D15	Each wire within its current rating at its continuous current	0 over; worst drop 48 mV	0 over	PASS
P-12	PWR-D07, IF-06	Key switch: high-side P-MOSFET, soft start, off without the key gate-drain (Miller) soft start: a gate-source RC gave 24 A (CR-13)	inrush 5.6 A, rail up in 2.5 ms	high side; soft start; fails off	PASS
P-13	IF-06, MCN-D54	Key out → rail < 0.5 V within 100 ms active bleeder Q3/R9, a pulse only: a shorted Q2 cannot burn it (FM-22 is found by the rail check)	58 ms (no bleeder: 956 ms)	≤ 100 ms	PASS
P-14	PWR-D08, IF-06	Rail divided to the helm's second voltage input; arming refused below 9.0 V	10 k / 1 k → × 11.0; 9.0 V → 0.82 V at ADC 15	BATT2_VOLT_MULT 11.0; ≤ 3.3 V	PASS
P-15	PWR-D09, PWR-D10	Rail off at power-up without the key; helm powered before the key	R3 holds Q2 off; FC and buck on VBAT_S (before Q2)	-	PASS
P-16	PWR-D11, IF-08	Mission computer: own 5.1 V ≥ 3 A buck, 7-14 V in; stays ≥ 4.85 V on a thrust step set-point and step response on the bench (IF-08 test)	KC-09 buck; 470 μF + TVS at the Pi; input ≥ 7.3 V at 70 W	5.1 V ± 2%	TEST
P-17	PWR-D16	No exposed live conductor outside the box	motor phases in IP68 connectors (caps when unmated); mast and switch leads in IP67 parts	none	PASS
P-18	PWR-009, PWR-D17, PWR-D18	Endurance at cruise, with this design's losses power.csv holds 11.1 W: add 0.24 W (CR-12)	11.34 W → 114 min on 21.6 Wh	≥ 40 min, margin ≥ 30%	PASS
P-19	PWR-010, PWR-D19, MCP-D04	Box temperature, 30 °C in sun, 60 min DUPLO plate shades 47% of the lid; test V-10	44 °C (3.6 W inside, 7.5 W sun)	≤ 50 °C	PASS
P-20	PWR-D19	Battery ≥ 30 mm from the ESCs box layout checked for overlaps and gland space	42 mm	≥ 30 mm	PASS
P-21	IF-06	Switch MOSFETs at the power-limited current box at 50 °C, R_DS(on) × 2.25 for low gate drive and heat	Tj 78 °C with a clip-on sink (127 °C without)	< 150 °C	PASS

8.3 Helm and mast

ID	Requirement	Check	Design value	Limit	Status
H-01	HLM-D02, HLM-D37	Flight controller resources used	SERIAL1 Pi, SERIAL3 GPS; UART4/5 free (ELRS spare); outputs 1, 4, 12; ADC 10, 11, 15; I ² C1	≥ 2 UARTs, 3 DShot, 2 ADC	PASS
H-02	HLM-D44, KF-04	One MAVLink path; wireless board not fitted	SERIAL1 only; SERIAL6 disabled	-	PASS
H-03	IF-05	ESC signals: output 1 left, 4 right, twisted pair ≤ 150 mm, no ESC power to the FC	TP1/TP2 150 mm; BEC leads unconnected	-	PASS

ID	Requirement	Check	Design value	Limit	Status
H-04	IF-04	UART to the Pi: 3.3 V both ends, crossed, common ground	Pi TXD → FC RX1, FC TX1 → Pi RXD, GND	no level shifting	PASS
H-05	IF-22, HLM-D35	GNSS on SERIAL3, compass on I ² C, through the mast connector confirm the module's pull-ups and the bus speed on the bench	0.9 m of cable; rise 0.56 μ s (4.7 k) / 0.26 μ s (2.2 k)	$\leq 1 \mu$ s (100 kHz), $\leq 0.3 \mu$ s (400 kHz)	TEST
H-06	IF-09, TBC-08	Beacon data level: 3.3 V data into the LED ring closes TBC-08 by design; confirm on the bench	with D6: V _{IH} margin ≥ 0.15 V (without: -0.34 V)	margin > 0	PASS

8.4 Mission computer

ID	Requirement	Check	Design value	Limit	Status
M-01	MCP-D03, FS-010	Moisture sensor read by GPIO trips with 0.8% of the comb wet	GPIO17 (software default); trips below 10.7 k Ω ; wet comb $\approx 82 \Omega$	GPIO; wet detected	PASS
M-02	MCP-D26, KC-12	DS18B20 at the box top on 1-Wire	GPIO4, 4.7 k Ω to 3.3 V	GPIO4, 4.7 k	PASS
M-03	MCP-D05	Mission computer power	1.8 W average (budget), 3 W peak; buck loss 0.2 W	≤ 2.0 W avg	PASS

8.5 Mission Control

ID	Requirement	Check	Design value	Limit	Status
C-01	IF-12, MCN-D02	Panel GPIOs agree with the software (boaty/mcn/panel.py)	buttons GO 17, HOME 27, STOP 22, TALK 24; LEDs 5, 6, 13, 19	identical	PASS
C-02	MCN-D01, MC-001	Mission Control on the power bank	9.7 h at 6.5 W	≥ 2.5 h	PASS
C-03	KC-16, MCN-1	Pi 5 USB current (usb_max_current_enable=1) phone on the tether as a USB SDP (≤ 0.5 A)	1.50 A	≤ 1.6 A	PASS

8.6 Safety

ID	Requirement	Check	Design value	Limit	Status
S-01	SAF-001, SAF-002	Helm independent of the mission computer and Mission Control	own supply from the bus; UART only link; ESC signals direct	-	PASS
S-02	FS-008	ESCs stop when commands stop V-09 on the bench	AM32 0.5 s (KC-03)	≤ 1 s	TEST
S-03	CHD-003, MEC-012	Electronics behind an adult action; every cable entry through a gland	latched box; 4 $\times$ PG7: motors $\times 2$, mast, switch	-	PASS
S-04	MEC-013, HLM-D35	GNSS ≥ 150 mm from power wiring	252 mm (MDD E-01; motor leads run down and aft)	≥ 150 mm	PASS

8.7 Cost

ID	Requirement	Check	Design value	Limit	Status
X-01	CON-001, CR-08	Bill of materials within the £185 cap (boat and shore kit) SDR BOM £181; this design adds £65 (connectors, cables, boards)	£246 buying everything; £229 with workshop stock (wire, resistors, stripboard)	$\leq$ £185	FAIL CR-14

9. Proposed change requests and cross-discipline actions

Proposals only: nothing baselined changes until the owner accepts a CR at the CDR.

CR-12 Power limit and power budget

Finding	P-06: at BATT_WATT_MAX 70 W the ESCs see 7.17 V at the critical battery, below PWR-D22's 7.5 V and the AM32's 7.2 V. The SDR analysis left out the harness, fuse, connectors and the two switch MOSFETs (34 mΩ).
Proposal	BATT_WATT_MAX 60 W: 7.63 V at the ESCs and 1.85 N of thrust (ENV-002 needs 1.1 N). Update power.csv with the buck and PIB losses (11.34 W at cruise).
Impact	software/params/boaty-mk1.parm (SITL re-run of the battery scenarios); SSS-HLM HLM-D43; KCL KF-02 and section 5; docs/budgets/power.csv.

CR-13 Key-switch circuit (IF-06, KCL KC-07)

Finding	IF-06 and KC-07 specify a gate RC soft start (≈ 10 ms) and no rail discharge. Simulated, the gate RC gives a 24 A inrush in 2.6 ms, and without a discharge path the ESC capacitors hold the rail above 0.5 V for 956 ms, failing IF-06's own test (< 0.5 V within 100 ms).
Proposal	Replace the text with this circuit: gate-to-drain (Miller) soft start, 470 nF; active bleeder (Q3, 10 Ω) pulsed by the key-out edge. Add the same soft start to the main switch (Q1). Results: inrush 5.6 A, key-out 58 ms.
Impact	ICD IF-06; KCL KC-07; FMEA FM-22 (the bleeder is a pulse, so a shorted Q2 cannot burn it).

CR-14 Cost cap

Finding	X-01: the SDR BOM (£181) has no lines for connectors, cables, board parts or wire. This design adds £65 (£48 if wire, resistors, stripboard and crimps come from stock): £229 against the £185 cap. The mechanical design adds stock items too (MDD O-3).
Largest items	M12 A 8-pole socket £6; Silicone wire 16/18/20/26 AWG, heat-shrink, tinned copper 1.5 mm ² for PIB bus strips £6; M12 A 8-pole plug £5; Cables: 8-core 0.25 mm ² 1 m, 3-core 0.75 mm ² 1.5 m, 2-core 0.5 m £5; IP68 3-pole socket £4; IP68 3-pole plug £4; IP68 3-pole socket £4; IP68 3-pole plug £4
Options	(a) Raise the cap to £240 (boat and shore kit). (b) Replace the M12 mast connector with a cheaper sealed 8-pin ($\approx -£6$) and the IP68 pod connectors with a cheaper keyed pair ($\approx -£8$), after the TBC-12 current check. (c) Hard-wire the pods ($-£16$), losing the tool-free pod swap (MEC-008). The owner decides; (a) with (b) is recommended.
Impact	SRS CON-001 (via CR-08's price rule); ADD and KCL BOM.

9.1 Actions on the mechanical design (MDD Issue B)

Action	Why
Handle part: mounting holes for SW1 (12 mm, with the cover's anti-rotation tab) and the J40 M12 socket (PG9 / 16 mm rear mount) on the uprights	PWR-004 switch outside the box; tool-free mast removal with its cable
Box: printed mezzanine plate on four stand-offs at 30 mm; reed-switch bracket bonded to the aft wall under the key dock	the box layout (Figure 4)
Stern segments: a clip holding each IP68 socket and its cap on the deck	PWR-D16: no exposed live contact
Pod strut: 4 mm groove for the motor lead, potted	MDD O-6
Masthead: a floor under the GNSS board with a cable gland-style exit into the tube	strain relief (IF-22)
Mass: the MDD mass table lacks the PIB, MIB, M12 and pod connectors and cables (estimate 60-100 g, to be weighed)	feeds CR-09

9.2 TBCs

TBC	Status
TBC-08 LED data level	Closed by design (D6); bench confirmation
TBC-06 motor current	Open: fuse and MOSFETs sized to the 8 A per ESC limit; bench measurement
TBC-12 pod connector	Open: requirement ≥ 10 A, keyed, IP68; candidate named in the BOM
TBC-13 box	Open (MDD): the layout fits the modelled box's 197 × 127 × 83 mm inside

10. Verification on the bench

ID	Test	Pass
E-V1	Power up with the key out: rail < 0.5 V; key in: rail = battery – 0.2 V; key out: < 0.5 V within 100 ms (scope)	IF-06
E-V2	Inrush on key-in and main-switch-on (current probe)	≤ 8 A
E-V3	Arming refused with the key out (BATT2 < 9.0 V)	V-13
E-V4	Calibrate BATT_VOLT_MULT and BATT_AMP_PERVLT at three loads	$\pm 1\%$ V, $\pm 5\%$ I
E-V5	Buck set-point and a full reverse-to-forward thrust step on a scope, at 12.6 V and 9.0 V	IF-08
E-V6	Beacon: every pattern shows with 3.3 V data through the full cable	TBC-08
E-V7	Compass reads over the mast cable at the helm's bus speed; motor-interference calibration	IF-22
E-V8	Moisture: a wet cloth across the comb → B6 RTL	FS-010
E-V9	ESC signal loss → stop ≤ 1 s	V-09
E-V10	Box 60 min at 30 °C in sun, Pi throttled flag = 0	V-10
E-V11	Every board: continuity against the placement list before power; then the smoke test on a current-limited supply	-

10.1 Open items

ID	Item
EO-1	Owner decisions on CR-12, CR-13, CR-14
EO-2	Confirm the PDB's battery-output pads are after its current sensor (V-01); if not, move the buck and key feed
EO-3	Confirm the candidate P-MOSFET's $R_{DS(on)} \leq 6$ m Ω from its datasheet before ordering
EO-4	Choose the pod connector (TBC-12) and the box (TBC-13)
EO-5	Rig L2 (iron bird) is this design on a board: build it first

Appendix A. Electrical parts

Ref	Value	Board	Part / candidate	£
BT1	3S1P 18650	PACK	KC-05 salvaged 18650 ×3	0.00
U1	3S BMS 20-25 A	PACK	KC-06 class	3.50
F1	20 A ATO	PACK		1.50
J1	XT60 (sockets)	PACK	XT60H-F	0.60
J2	JST-XH 4	PACK	XHP-4	0.20
J3	XT60 (pins)	BOX	XT60H-M	0.60
U2	SpeedyBee F405 WING APP	BOX	KC-01 SpeedyBee F405 WING APP	45.00
U3	AM32 20 A	BOX	KC-03 AM32 6S 20 A	11.00
U4	AM32 20 A	BOX	KC-03 AM32 6S 20 A	11.00
C7	470µF	BOX		0.40
C8	470µF	BOX		0.40
U5	Buck 5.1 V 3 A	BOX	KC-09 class (the KCL's £3 module); bench-checked to IF-08. Upgrade: Po	3.00
Q1	P-MOSFET	PIB	candidate Infineon IPP80P03P4L-04 (−30 V, R_DS(on) ≤ 6 mΩ at −10 V ass	1.80
Q2	P-MOSFET	PIB	candidate Infineon IPP80P03P4L-04 (−30 V, R_DS(on) ≤ 6 mΩ at −10 V ass	1.80
Q3	IRLZ44N	PIB	IRLZ44NPbF	1.00
R1	10k	PIB		0.02
R2	3k3	PIB		0.02
C1	470nF	PIB		0.05
D1	BZX85C15	PIB	BZX85C15	0.10
D2	P6KE18A	PIB	P6KE18A	0.30
R3	10k	PIB		0.02
R4	3k3	PIB		0.02
C2	470nF	PIB		0.05
D3	BZX85C15	PIB	BZX85C15	0.10
C3	1µF	PIB		0.05
R5	100k	PIB		0.02
D4	BZX85C10	PIB	BZX85C10	0.10
R9	10R	PIB		0.20
D5	P6KE18A	PIB	P6KE18A	0.30
R6	10k	PIB		0.02
R7	1k	PIB		0.02
C4	100nF	PIB		0.05
F2	PTC 1.1 A	PIB	RXEF110	0.30
D6	1N4001	PIB	1N4001	0.10
R8	330R	PIB		0.02
J11	KK254 2	PIB	Molex 22-27-2021	0.30
J12	KK254 3	PIB	Molex 22-27-2031	0.30
J14	KK254 8	PIB	Molex 22-27-2081	0.30

Ref	Value	Board	Part / candidate	£
J15	KK254 6	PIB	Molex 22-27-2061	0.30
SW1	IP67 toggle + cover	HANDLE	IP67 sealed miniature toggle, ≥ 0.5 A 24 V DC, with flip cover	4.00
SW2	Reed NO	BOX	normally-open glass reed, 10-20 AT (e.g. Littelfuse 59140 housed type)	1.50
J40	M12 A 8-pole socket	HANDLE	M12 A-coded 8-pole panel socket, rear mount, IP67	6.00
P40	M12 A 8-pole plug	MASTHEAD	M12 A-coded 8-pole cable plug, IP67	5.00
U9	M10 GNSS + compass	MASTHEAD	KC-02 M10 + QMC5883L	14.00
U10	WS2812B ring $\times 8$	MASTHEAD	KC-14	3.00
J50	IP68 3-pole socket	DECK_P	TBC-12: keyed 3-pole IP68, ≥ 10 A (e.g. Weipu SP13 3-pole; confirm rat	4.00
P50	IP68 3-pole plug	POD_P	as the socket	4.00
M1	2205 outrunner	POD_P	KC-04	6.00
J51	IP68 3-pole socket	DECK_S	TBC-12: keyed 3-pole IP68, ≥ 10 A (e.g. Weipu SP13 3-pole; confirm rat	4.00
P51	IP68 3-pole plug	POD_S	as the socket	4.00
M2	2205 outrunner	POD_S	KC-04	6.00
J30	Micro-Fit 2 plug	BOX	Molex 43025-0200	0.50
J31	Micro-Fit 2 header	MIB	Molex 43650-0200	0.60
D10	P6KE6.8A	MIB	P6KE6.8A	0.30
C10	470 μ F	MIB		0.30
C11	10 μ F	MIB		0.05
J32	2 $\times$ 20 socket	MIB	2 $\times$ 20 2.54 mm female, stacking	1.00
U6	Raspberry Pi Zero 2 W	BOX	KC-10	15.00
U7	OV5647 camera	BOX	KC-11	8.00
J33	KK254 3	MIB	Molex 22-27-2031	0.30
J34	KK254 4	MIB	Molex 22-27-2041	0.30
J35	KK254 2	MIB	Molex 22-27-2021	0.30
R10	4k7	MIB		0.02
R11	1k	MIB		0.02
R12	100k	MIB		0.02
C12	100nF	MIB		0.05
U8	DS18B20	BOX	KC-12	2.00
S2	Moisture comb	BOX	nickel-plated comb board (rain-sensor type, comb only)	1.00
U50	Raspberry Pi 5	MC	KC-16 (owned)	0.00
J60	2 $\times$ 20 IDC header	PNL		0.80
BT50	USB-C PD bank 20 Ah	MC	20,000 mAh USB-C PD, 5 V 3 A (owned or bought)	0.00
U51	Wi-Fi adapter	MC		12.00
U52	USB microphone	MC		4.00
U53	USB speaker	MC		4.00
U54	Phone (tether)	MC		0.00
SW61	GO button	MC	KC-18	1.50

Ref	Value	Board	Part / candidate	£
R61	10k	PNL		0.02
R62	1k	PNL		0.02
C61	100nF	PNL		0.05
Q61	2N7000	PNL	2N7000	0.10
R63	100R	PNL		0.02
R64	100k	PNL		0.02
SW62	HOME button	MC	KC-18	1.50
R71	10k	PNL		0.02
R72	1k	PNL		0.02
C62	100nF	PNL		0.05
Q62	2N7000	PNL	2N7000	0.10
R73	100R	PNL		0.02
R74	100k	PNL		0.02
SW63	STOP button	MC	KC-18	1.50
R81	10k	PNL		0.02
R82	1k	PNL		0.02
C63	100nF	PNL		0.05
Q63	2N7000	PNL	2N7000	0.10
R83	100R	PNL		0.02
R84	100k	PNL		0.02
SW64	TALK button	MC	KC-18	1.50
R91	10k	PNL		0.02
R92	1k	PNL		0.02
C64	100nF	PNL		0.05
Q64	2N7000	PNL	2N7000	0.10
R93	100R	PNL		0.02
R94	100k	PNL		0.02

Appendix B. Pin maps

B.1 Flight controller (U2)

Pin	Function	Net
BAT+		BUS_POS
BAT-		GND
VBO+	battery out after the current sensor	VBAT_S
VBO-		GND
S1	PWM1 / DShot, BIDIR	ESC_L
G1		GND
S4	PWM4 / DShot, BIDIR	ESC_R
G4		GND
TX1	USART1 TX (SERIAL1)	MCP_RX
RX1	USART1 RX (FT pin)	MCP_TX
GU1		GND
TX3	USART3 TX (SERIAL3, GPS)	GPS_FC_TX
RX3	USART3 RX	GPS_FC_RX
4V5	GPS port 4.5 V	GPS_4V5
GGPS		GND
SCL	I2C1 SCL	I2C_SCL
SDA	I2C1 SDA	I2C_SDA
LED	PWM12 / NeoPixel	LED_DATA_FC
5V	LED pad 5 V (5 V BEC)	LED_5V_FC
AIR	AIRSPD pad, ADC pin 15	RAIL_ADC
GAIR		GND

B.2 Pi Zero 2 W (U6)

Pin	Function	Net
1	3V3	3V3_MCP
2	5V	5V_MCP
4	5V	5V_MCP
6	GND	GND
7	GPIO4 1-Wire	W1_DQ
8	GPIO14 TXD	MCP_TX
9	GND	GND
10	GPIO15 RXD	MCP_RX
11	GPIO17 moisture	MOIST_GPIO
14	GND	GND
CSI	camera ribbon	CSI

B.3 Raspberry Pi 5 (U50)

Pin	Function	Net
1	3V3	PI5_3V3
2	5V	PI5_5V

Pin	Function	Net
6	GND	PI5_GND
9	GND	PI5_GND
14	GND	PI5_GND
11	GPIO17 GO	BTN_GO
29	GPIO5 GO LED	LED_GO
13	GPIO27 HOME	BTN_HOME
31	GPIO6 HOME LED	LED_HOME
15	GPIO22 STOP	BTN_STOP
33	GPIO13 STOP LED	LED_STOP
18	GPIO24 TALK	BTN_TALK
35	GPIO19 TALK LED	LED_TALK
USBC	USB-C power in	PI5_5V_IN
USB1		USB_WIFI
USB2		USB_MIC
USB3		USB_SPK
USB4		USB_PHONE